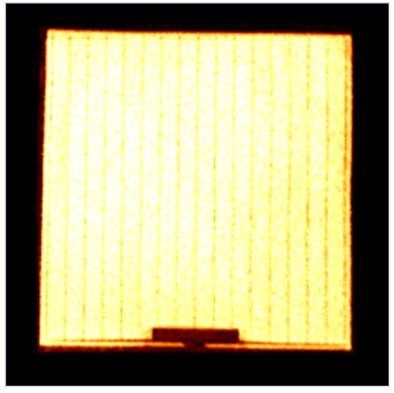
Reference | image 0021 | 20% PCM / NPL default

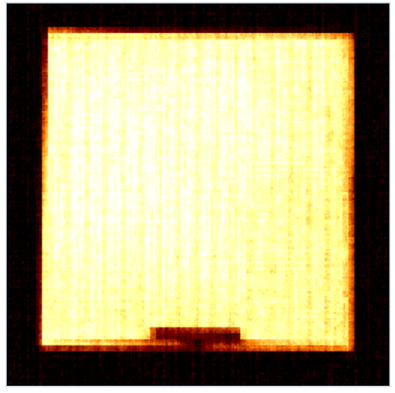
Review thin-line contrast, especially U-TV C60 against GS-PnP and old U-TV.

Ground truth



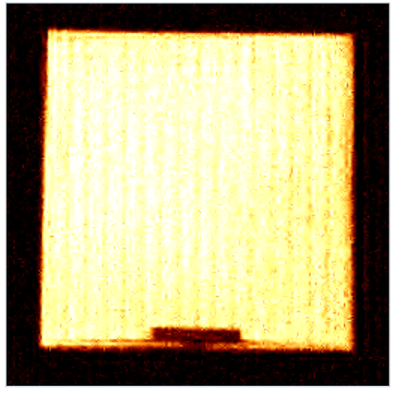
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



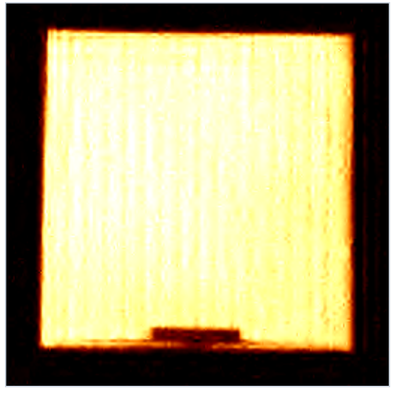
PSNR 24.47 | SSIM 0.509
HaarPSI 0.564 | MSE 3.58e-03

SPGL1-BP



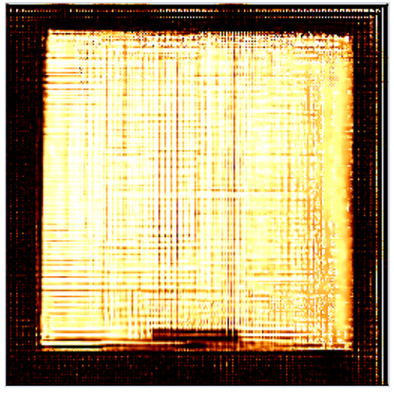
PSNR 24.04 | SSIM 0.453
HaarPSI 0.617 | MSE 3.95e-03

SPGL1-BPDN



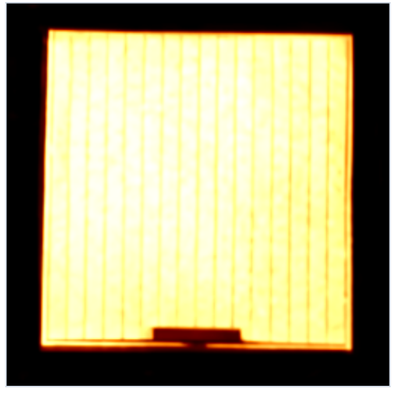
PSNR 26.06 | SSIM 0.580
HaarPSI 0.591 | MSE 2.48e-03

PnP-ADMM DRUNet



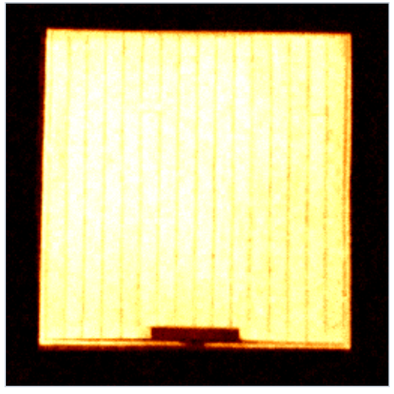
PSNR 10.38 | SSIM 0.186
HaarPSI 0.391 | MSE 9.16e-02

PnP-ADMM DRUNet tuned



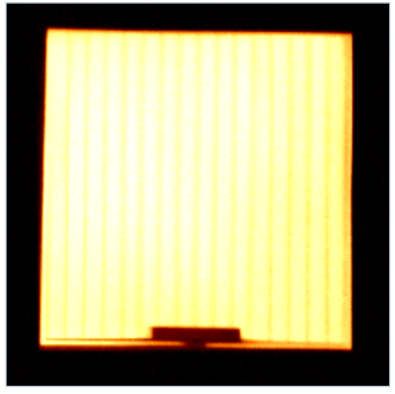
PSNR 31.07 | SSIM 0.847
HaarPSI 0.822 | MSE 7.82e-04

GS-PnP GS-DRUNet



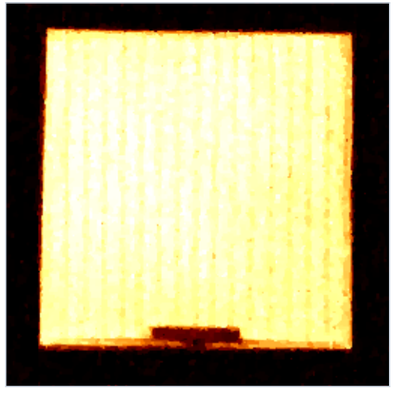
PSNR 29.11 | SSIM 0.680
HaarPSI 0.799 | MSE 1.23e-03

GS-PnP GS-DRUNet tuned



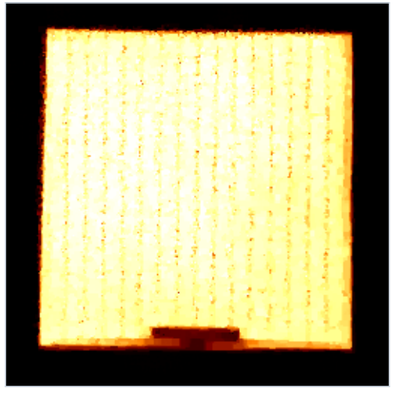
PSNR 29.33 | SSIM 0.748
HaarPSI 0.752 | MSE 1.17e-03

TV tuned (0.011)



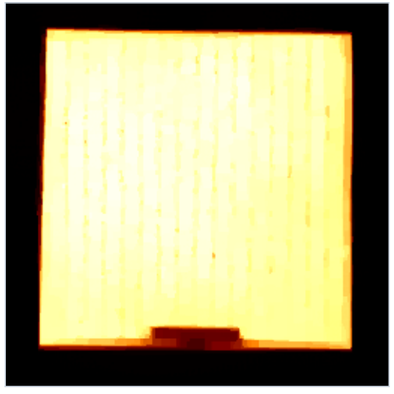
PSNR 27.48 | SSIM 0.642
HaarPSI 0.692 | MSE 1.79e-03

Old clean U-TV



PSNR 26.32 | SSIM 0.719
HaarPSI 0.711 | MSE 2.34e-03

New U-TV C60

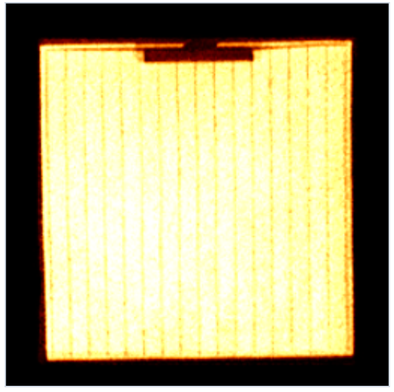


PSNR 27.83 | SSIM 0.695
HaarPSI 0.574 | MSE 1.65e-03

Reference | image 0066 | 20% PCM / NPL default

Review thin-line contrast, especially U-TV C60 against GS-PnP and old U-TV.

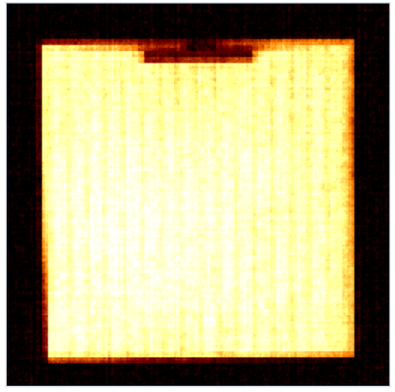
Ground truth



PSNR inf | SSIM 1.000

HaarPSI 1.000 | MSE 0

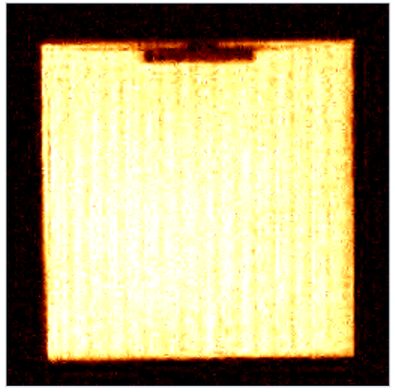
Zero-filled



PSNR 24.59 | SSIM 0.516

HaarPSI 0.544 | MSE 3.48e-03

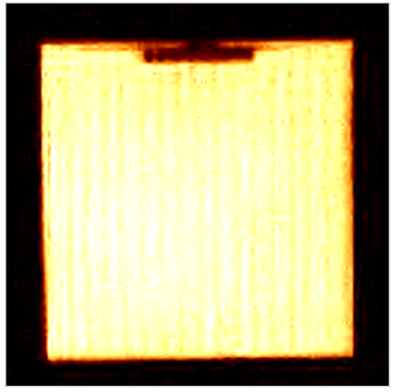
SPGL1-BP



PSNR 24.15 | SSIM 0.453

HaarPSI 0.616 | MSE 3.84e-03

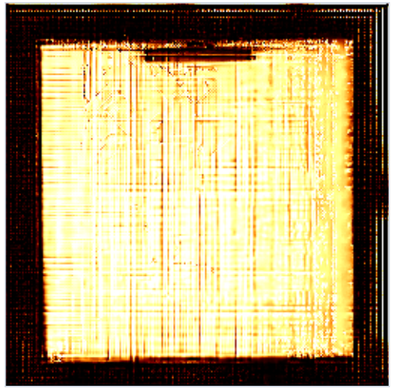
SPGL1-BPDN



PSNR 26.25 | SSIM 0.579

HaarPSI 0.599 | MSE 2.37e-03

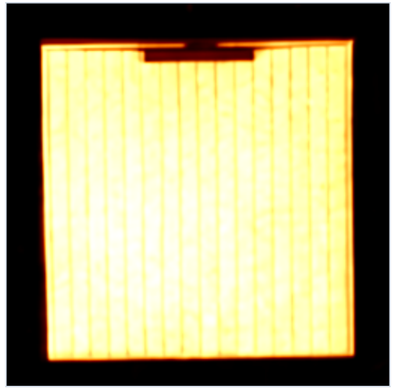
PnP-ADMM DRUNet



PSNR 12.70 | SSIM 0.227

HaarPSI 0.358 | MSE 5.36e-02

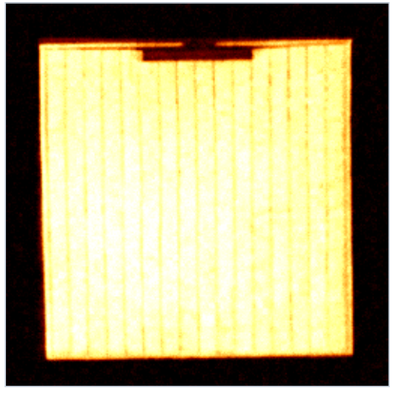
PnP-ADMM DRUNet tuned



PSNR 31.05 | SSIM 0.846

HaarPSI 0.814 | MSE 7.85e-04

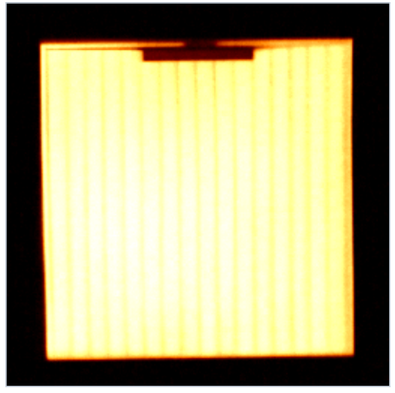
GS-PnP GS-DRUNet



PSNR 29.07 | SSIM 0.680

HaarPSI 0.795 | MSE 1.24e-03

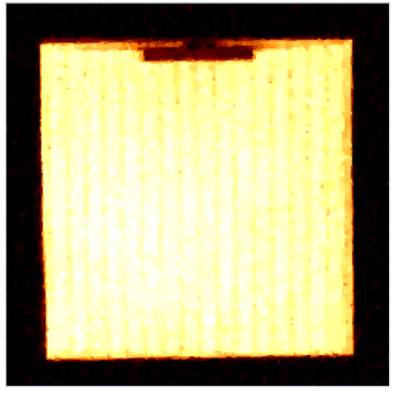
GS-PnP GS-DRUNet tuned



PSNR 29.38 | SSIM 0.754

HaarPSI 0.747 | MSE 1.15e-03

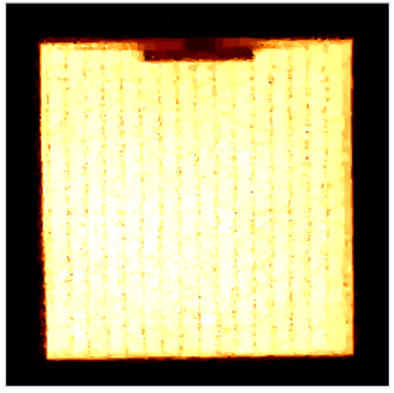
TV tuned (0.011)



PSNR 27.54 | SSIM 0.650

HaarPSI 0.684 | MSE 1.76e-03

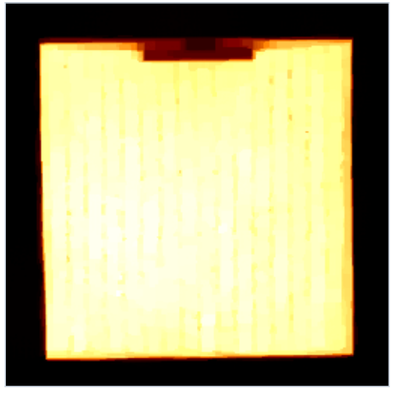
Old clean U-TV



PSNR 26.49 | SSIM 0.728

HaarPSI 0.725 | MSE 2.24e-03

New U-TV C60



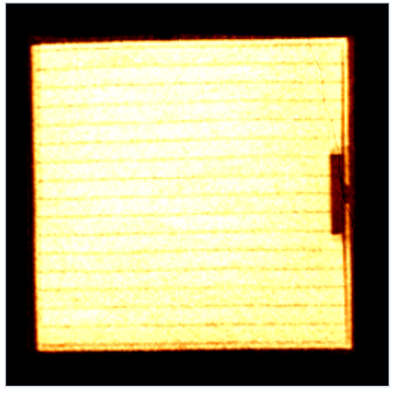
PSNR 27.67 | SSIM 0.688

HaarPSI 0.568 | MSE 1.71e-03

Reference | image 0188 | 20% PCM / NPL default

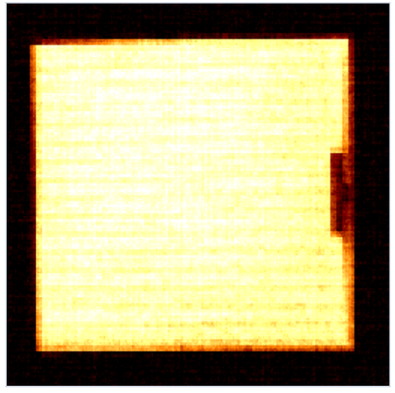
Review thin-line contrast, especially U-TV C60 against GS-PnP and old U-TV.

Ground truth



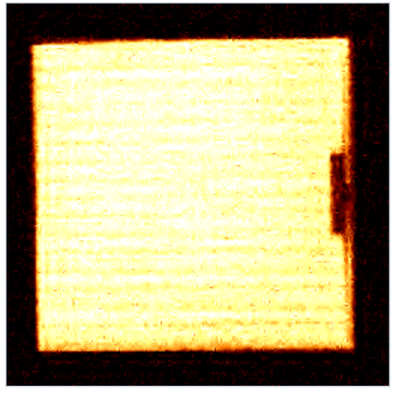
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



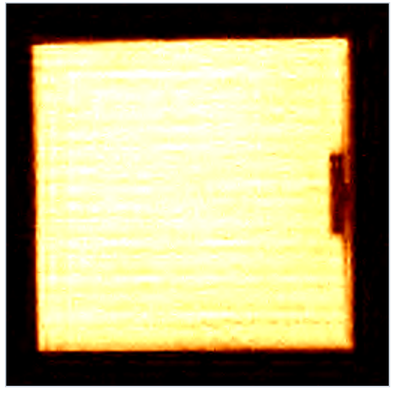
PSNR 24.51 | SSIM 0.510
HaarPSI 0.565 | MSE 3.54e-03

SPGL1-BP



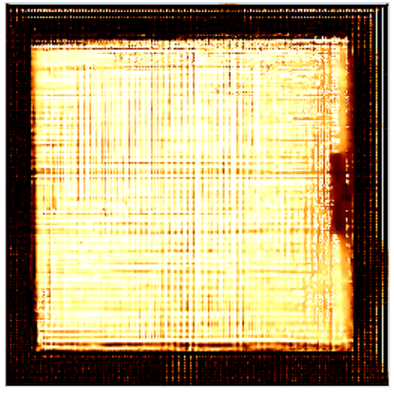
PSNR 24.01 | SSIM 0.445
HaarPSI 0.611 | MSE 3.97e-03

SPGL1-BPDN



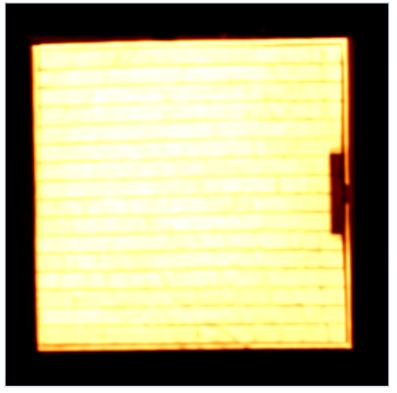
PSNR 26.02 | SSIM 0.573
HaarPSI 0.606 | MSE 2.50e-03

PnP-ADMM DRUNet



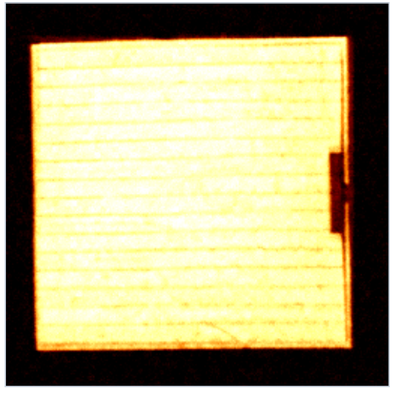
PSNR 11.19 | SSIM 0.186
HaarPSI 0.346 | MSE 7.61e-02

PnP-ADMM DRUNet tuned



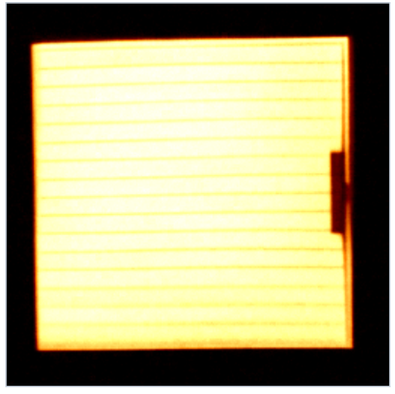
PSNR 30.84 | SSIM 0.838
HaarPSI 0.818 | MSE 8.25e-04

GS-PnP GS-DRUNet



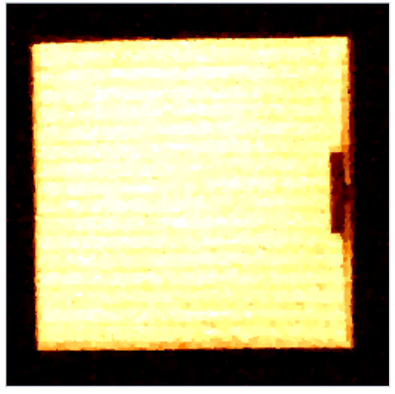
PSNR 28.70 | SSIM 0.664
HaarPSI 0.785 | MSE 1.35e-03

GS-PnP GS-DRUNet tuned



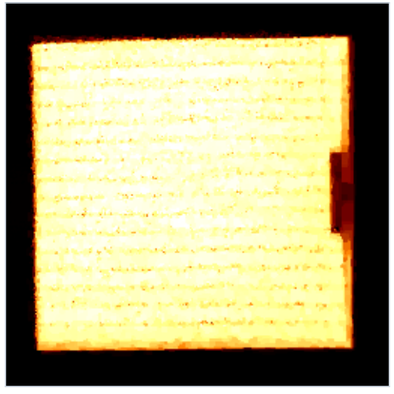
PSNR 29.84 | SSIM 0.780
HaarPSI 0.766 | MSE 1.04e-03

TV tuned (0.011)



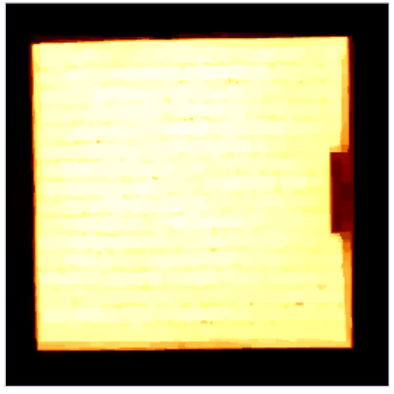
PSNR 27.35 | SSIM 0.643
HaarPSI 0.685 | MSE 1.84e-03

Old clean U-TV



PSNR 26.28 | SSIM 0.717
HaarPSI 0.711 | MSE 2.35e-03

New U-TV C60

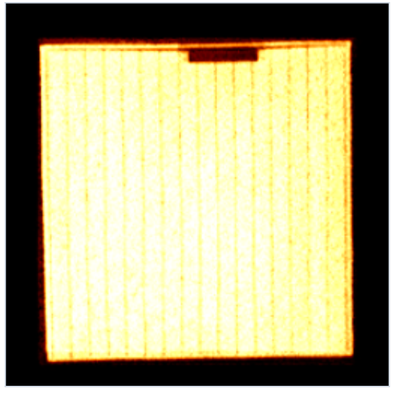


PSNR 27.74 | SSIM 0.700
HaarPSI 0.596 | MSE 1.68e-03

Reference | image 0727 | 20% PCM / NPL default

Review thin-line contrast, especially U-TV C60 against GS-PnP and old U-TV.

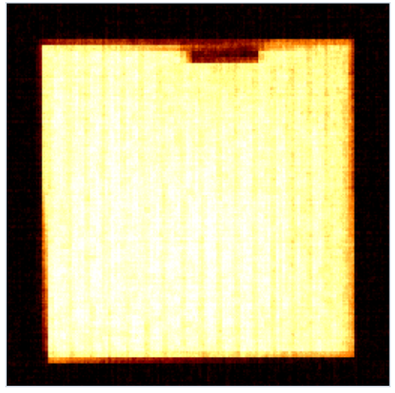
Ground truth



PSNR inf | SSIM 1.000

HaarPSI 1.000 | MSE 0

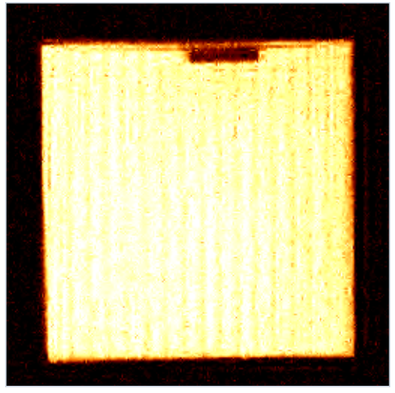
Zero-filled



PSNR 24.59 | SSIM 0.511

HaarPSI 0.574 | MSE 3.48e-03

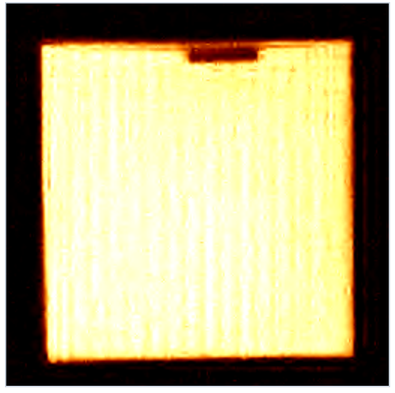
SPGL1-BP



PSNR 23.94 | SSIM 0.450

HaarPSI 0.616 | MSE 4.03e-03

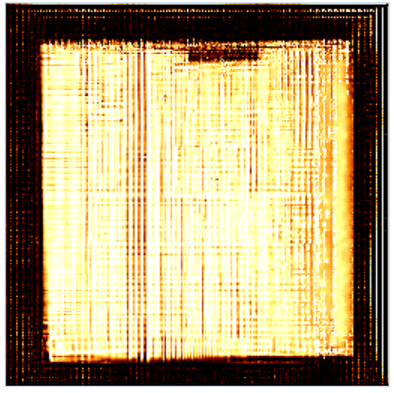
SPGL1-BPDN



PSNR 26.02 | SSIM 0.578

HaarPSI 0.607 | MSE 2.50e-03

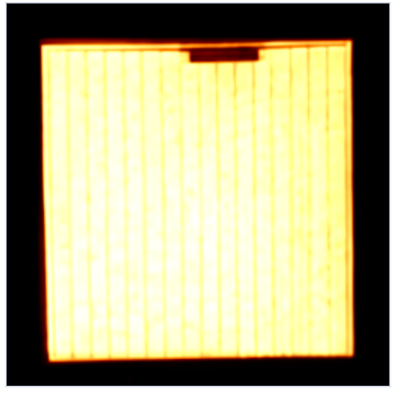
PnP-ADMM DRUNet



PSNR 9.36 | SSIM 0.172

HaarPSI 0.325 | MSE 1.16e-01

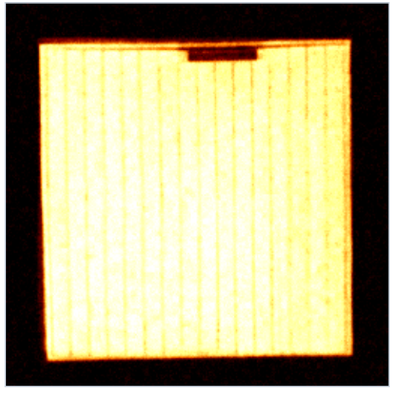
PnP-ADMM DRUNet tuned



PSNR 31.04 | SSIM 0.845

HaarPSI 0.828 | MSE 7.88e-04

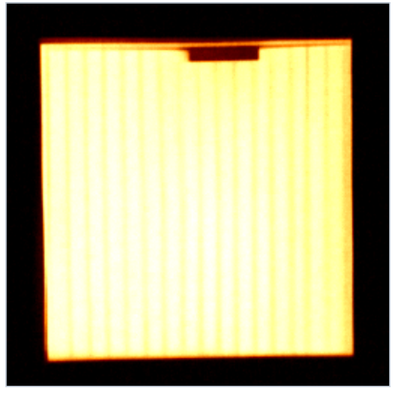
GS-PnP GS-DRUNet



PSNR 28.94 | SSIM 0.673

HaarPSI 0.794 | MSE 1.28e-03

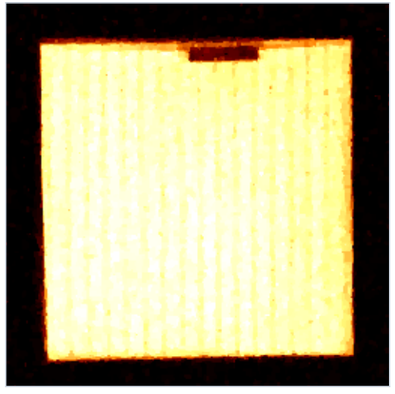
GS-PnP GS-DRUNet tuned



PSNR 29.02 | SSIM 0.736

HaarPSI 0.737 | MSE 1.25e-03

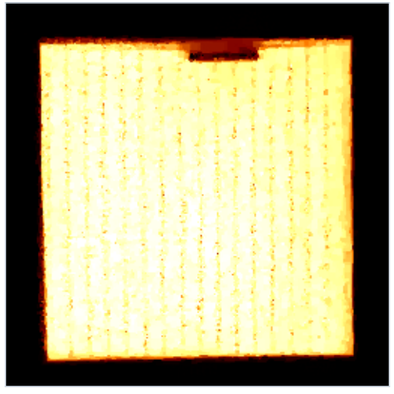
TV tuned (0.011)



PSNR 27.44 | SSIM 0.647

HaarPSI 0.677 | MSE 1.80e-03

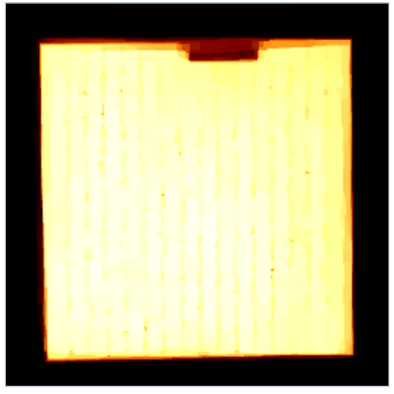
Old clean U-TV



PSNR 26.15 | SSIM 0.716

HaarPSI 0.708 | MSE 2.43e-03

New U-TV C60



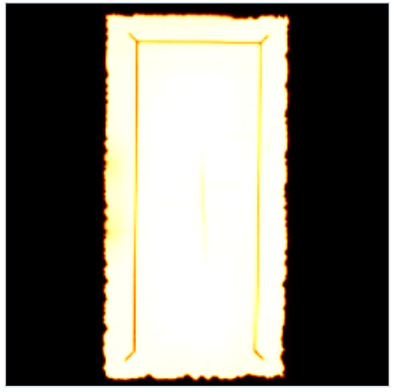
PSNR 27.83 | SSIM 0.698

HaarPSI 0.600 | MSE 1.65e-03

CIGS | image 0710 | 20% PCM / NPL default

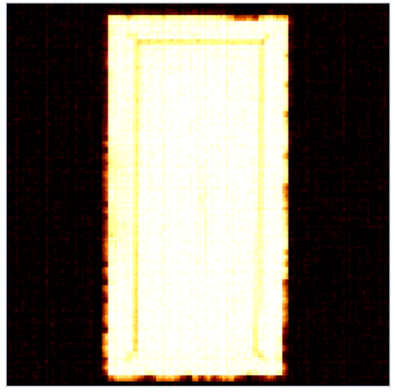
Review straight-line continuity and sharpness across reconstructions.

Ground truth



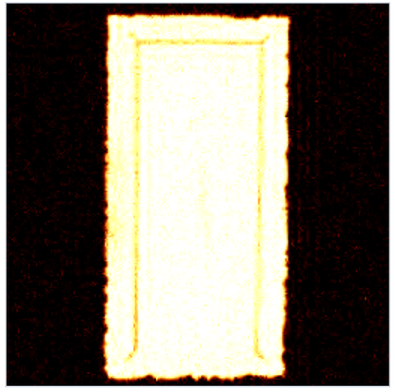
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



PSNR 24.80 | SSIM 0.411
HaarPSI 0.493 | MSE 3.31e-03

SPGL1-BP



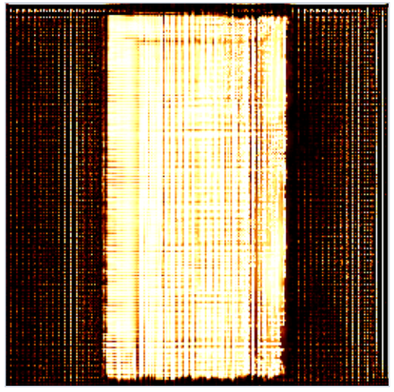
PSNR 25.75 | SSIM 0.402
HaarPSI 0.588 | MSE 2.66e-03

SPGL1-BPDN



PSNR 28.87 | SSIM 0.756
HaarPSI 0.587 | MSE 1.30e-03

PnP-ADMM DRUNet



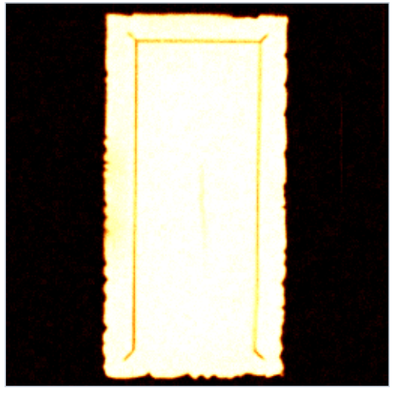
PSNR 8.09 | SSIM 0.102
HaarPSI 0.246 | MSE 1.55e-01

PnP-ADMM DRUNet tuned



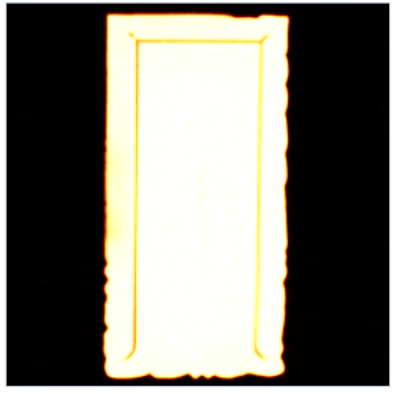
PSNR 37.25 | SSIM 0.985
HaarPSI 0.937 | MSE 1.88e-04

GS-PnP GS-DRUNet



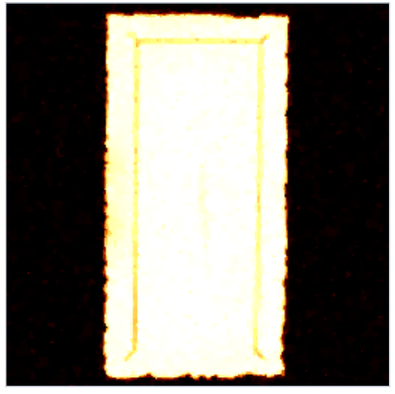
PSNR 31.21 | SSIM 0.613
HaarPSI 0.794 | MSE 7.57e-04

GS-PnP GS-DRUNet tuned



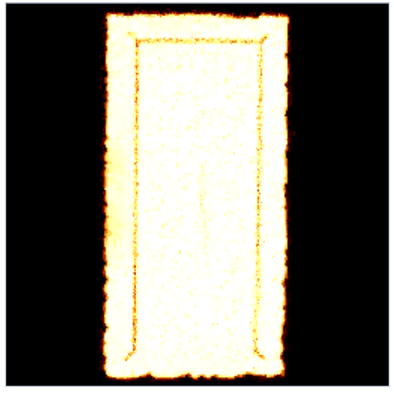
PSNR 33.85 | SSIM 0.893
HaarPSI 0.888 | MSE 4.12e-04

TV tuned (0.011)



PSNR 30.48 | SSIM 0.738
HaarPSI 0.801 | MSE 8.95e-04

Old clean U-TV



PSNR 28.52 | SSIM 0.794
HaarPSI 0.776 | MSE 1.41e-03

New U-TV C60

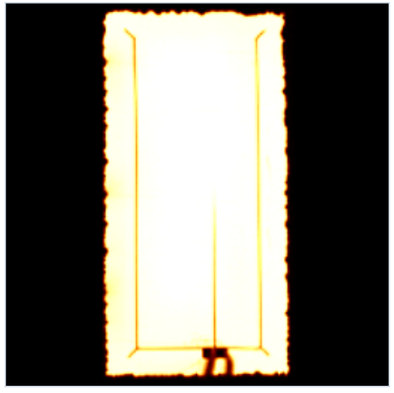


PSNR 32.60 | SSIM 0.957
HaarPSI 0.875 | MSE 5.50e-04

CIGS | image 0727 | 20% PCM / NPL default

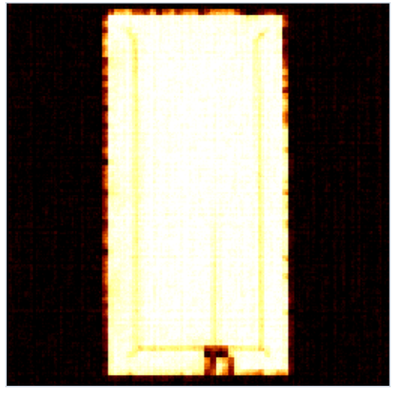
Review straight-line continuity and sharpness across reconstructions.

Ground truth



PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



PSNR 24.31 | SSIM 0.417
HaarPSI 0.489 | MSE 3.71e-03

SPGL1-BP



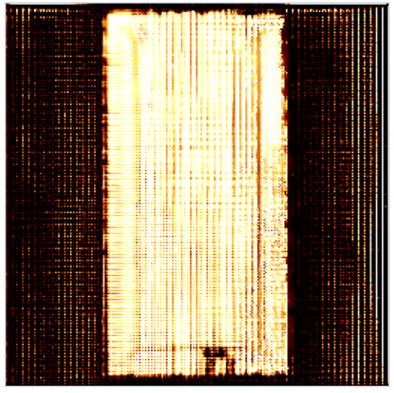
PSNR 25.19 | SSIM 0.390
HaarPSI 0.572 | MSE 3.03e-03

SPGL1-BPDN



PSNR 27.62 | SSIM 0.712
HaarPSI 0.555 | MSE 1.73e-03

PnP-ADMM DRUNet



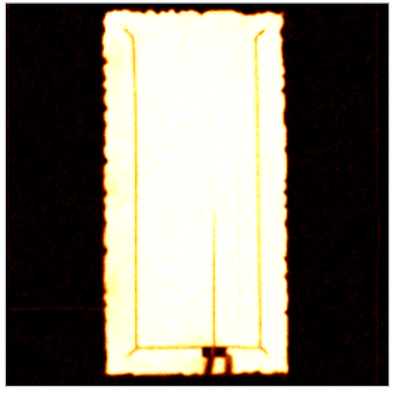
PSNR 8.18 | SSIM 0.113
HaarPSI 0.275 | MSE 1.52e-01

PnP-ADMM DRUNet tuned



PSNR 36.98 | SSIM 0.978
HaarPSI 0.927 | MSE 2.00e-04

GS-PnP GS-DRUNet



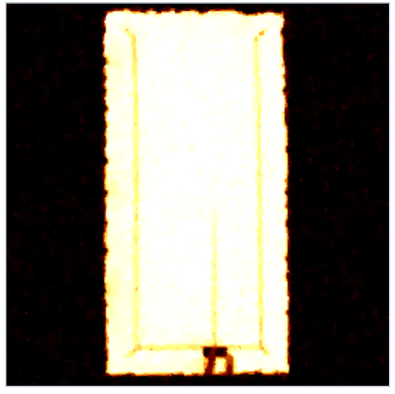
PSNR 30.80 | SSIM 0.604
HaarPSI 0.796 | MSE 8.32e-04

GS-PnP GS-DRUNet tuned



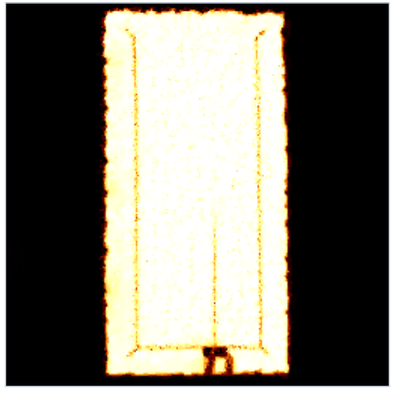
PSNR 33.82 | SSIM 0.894
HaarPSI 0.885 | MSE 4.15e-04

TV tuned (0.011)



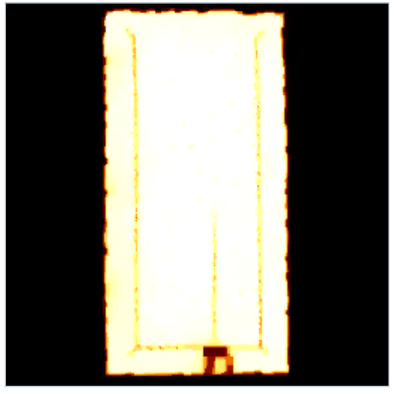
PSNR 30.12 | SSIM 0.726
HaarPSI 0.797 | MSE 9.73e-04

Old clean U-TV



PSNR 28.38 | SSIM 0.794
HaarPSI 0.774 | MSE 1.45e-03

New U-TV C60



PSNR 31.51 | SSIM 0.945
HaarPSI 0.856 | MSE 7.06e-04

CIGS | image 0805 | 20% PCM / NPL default

Review straight-line continuity and sharpness across reconstructions.

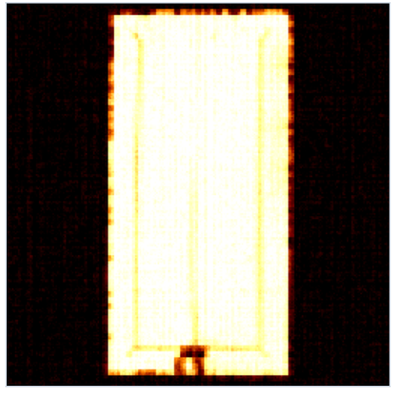
Ground truth



PSNR inf | SSIM 1.000

HaarPSI 1.000 | MSE 0

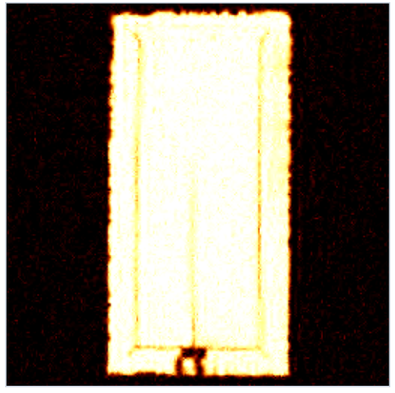
Zero-filled



PSNR 24.34 | SSIM 0.419

HaarPSI 0.500 | MSE 3.68e-03

SPGL1-BP



PSNR 25.37 | SSIM 0.398

HaarPSI 0.565 | MSE 2.90e-03

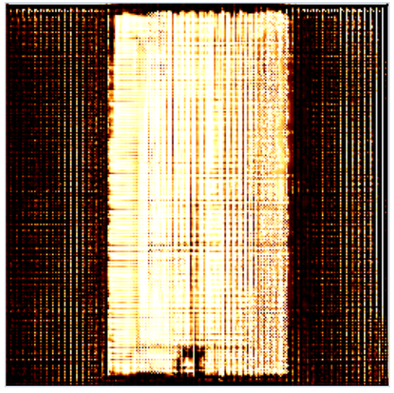
SPGL1-BPDN



PSNR 28.09 | SSIM 0.725

HaarPSI 0.554 | MSE 1.55e-03

PnP-ADMM DRUNet



PSNR 6.49 | SSIM 0.096

HaarPSI 0.288 | MSE 2.24e-01

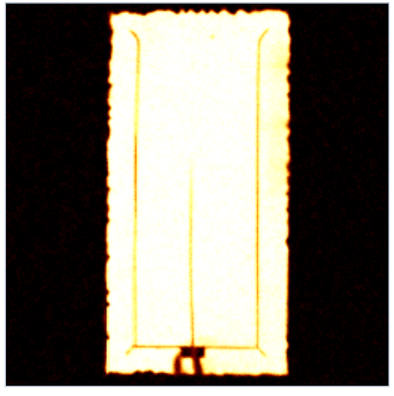
PnP-ADMM DRUNet tuned



PSNR 36.81 | SSIM 0.982

HaarPSI 0.926 | MSE 2.09e-04

GS-PnP GS-DRUNet



PSNR 31.18 | SSIM 0.626

HaarPSI 0.817 | MSE 7.62e-04

GS-PnP GS-DRUNet tuned



PSNR 32.77 | SSIM 0.890

HaarPSI 0.871 | MSE 5.28e-04

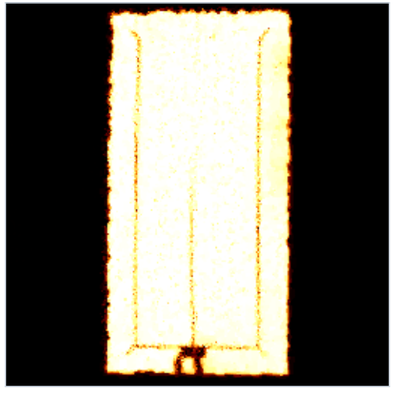
TV tuned (0.011)



PSNR 30.41 | SSIM 0.745

HaarPSI 0.807 | MSE 9.10e-04

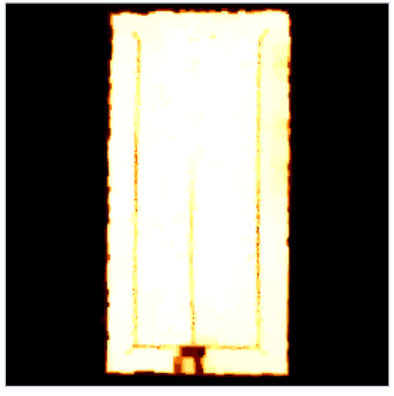
Old clean U-TV



PSNR 28.20 | SSIM 0.807

HaarPSI 0.786 | MSE 1.51e-03

New U-TV C60



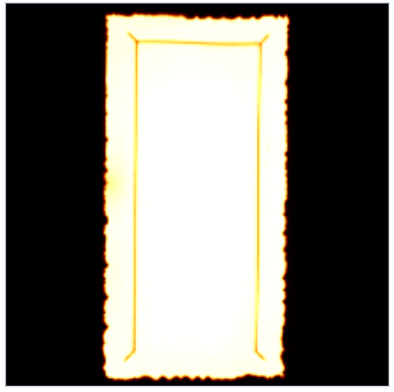
PSNR 31.65 | SSIM 0.951

HaarPSI 0.868 | MSE 6.85e-04

CIGS | image 0821 | 20% PCM / NPL default

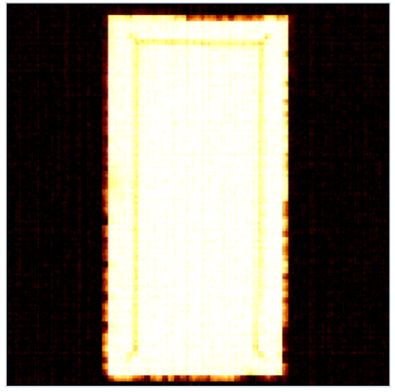
Review straight-line continuity and sharpness across reconstructions.

Ground truth



PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



PSNR 24.52 | SSIM 0.403
HaarPSI 0.492 | MSE 3.53e-03

SPGL1-BP



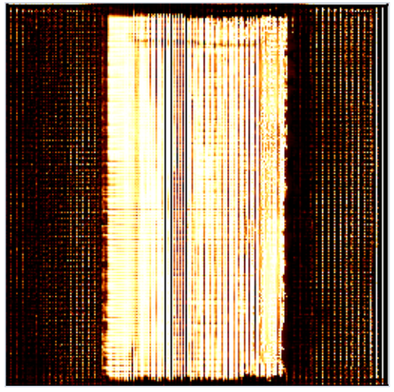
PSNR 25.43 | SSIM 0.387
HaarPSI 0.544 | MSE 2.86e-03

SPGL1-BPDN



PSNR 28.31 | SSIM 0.735
HaarPSI 0.536 | MSE 1.48e-03

PnP-ADMM DRUNet



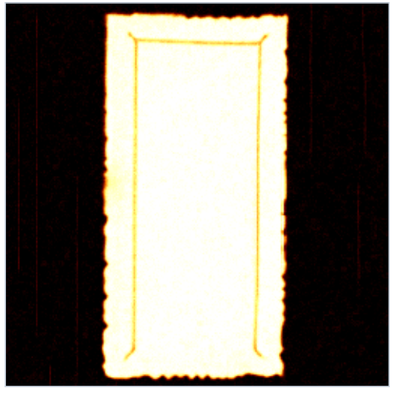
PSNR 5.63 | SSIM 0.099
HaarPSI 0.238 | MSE 2.73e-01

PnP-ADMM DRUNet tuned



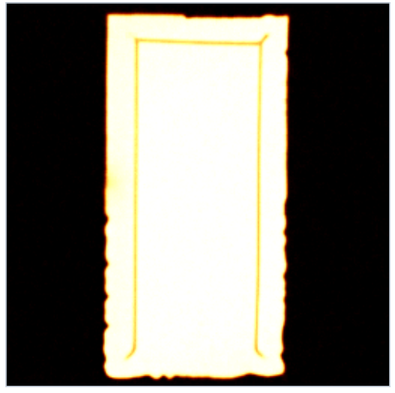
PSNR 37.80 | SSIM 0.987
HaarPSI 0.938 | MSE 1.66e-04

GS-PnP GS-DRUNet



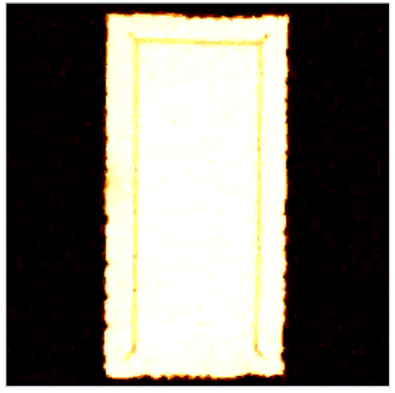
PSNR 30.63 | SSIM 0.595
HaarPSI 0.788 | MSE 8.65e-04

GS-PnP GS-DRUNet tuned



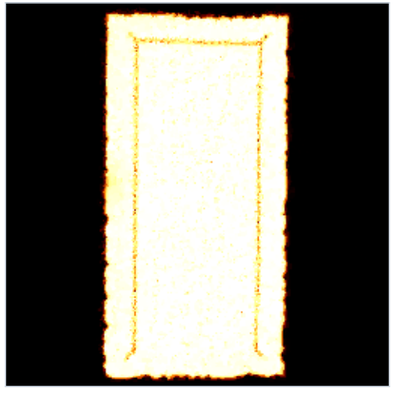
PSNR 33.81 | SSIM 0.896
HaarPSI 0.896 | MSE 4.16e-04

TV tuned (0.011)



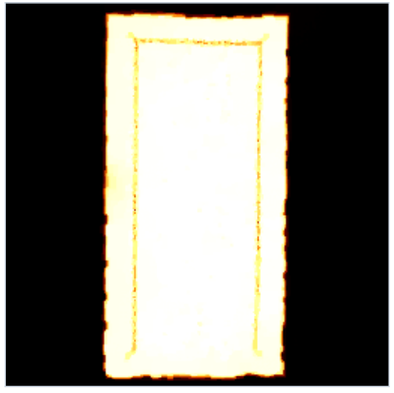
PSNR 30.95 | SSIM 0.741
HaarPSI 0.808 | MSE 8.04e-04

Old clean U-TV



PSNR 28.77 | SSIM 0.796
HaarPSI 0.797 | MSE 1.33e-03

New U-TV C60

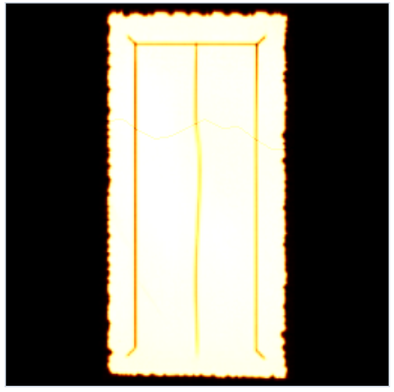


PSNR 32.53 | SSIM 0.960
HaarPSI 0.892 | MSE 5.59e-04

CIGS | image 0873 | 20% PCM / NPL default

Review straight-line continuity and sharpness across reconstructions.

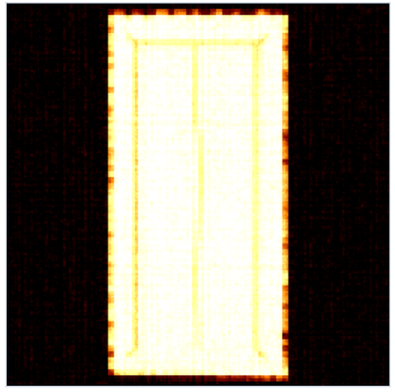
Ground truth



PSNR inf | SSIM 1.000

HaarPSI 1.000 | MSE 0

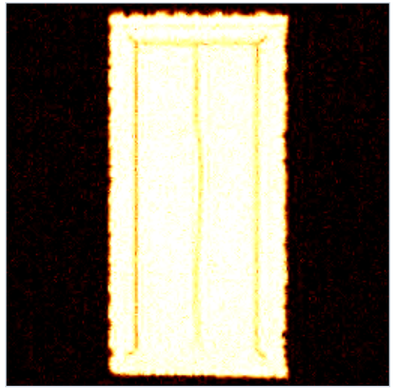
Zero-filled



PSNR 23.61 | SSIM 0.416

HaarPSI 0.458 | MSE 4.35e-03

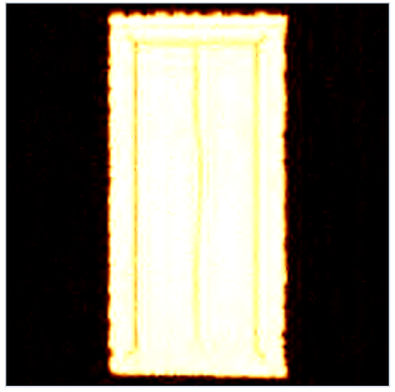
SPGL1-BP



PSNR 25.46 | SSIM 0.406

HaarPSI 0.579 | MSE 2.85e-03

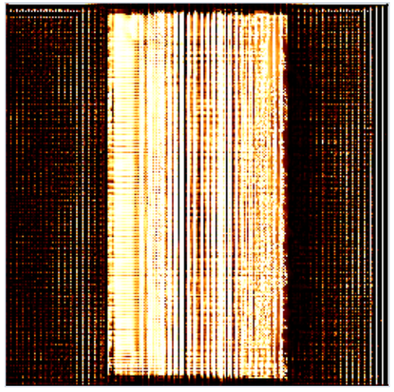
SPGL1-BPDN



PSNR 28.43 | SSIM 0.740

HaarPSI 0.591 | MSE 1.44e-03

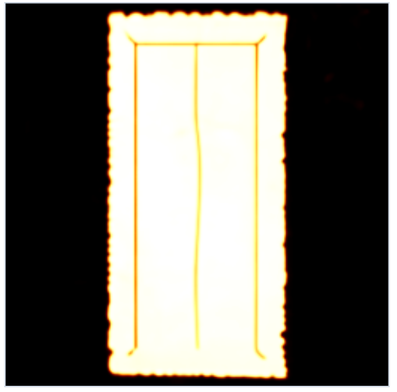
PnP-ADMM DRUNet



PSNR 3.34 | SSIM 0.088

HaarPSI 0.196 | MSE 4.64e-01

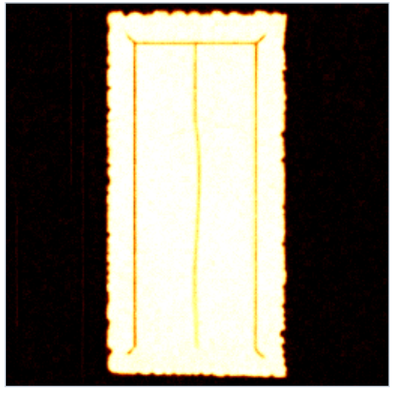
PnP-ADMM DRUNet tuned



PSNR 37.82 | SSIM 0.979

HaarPSI 0.946 | MSE 1.65e-04

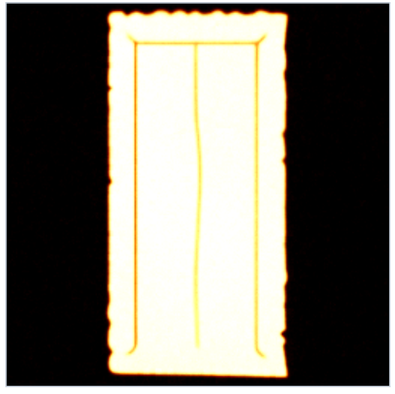
GS-PnP GS-DRUNet



PSNR 31.08 | SSIM 0.615

HaarPSI 0.803 | MSE 7.80e-04

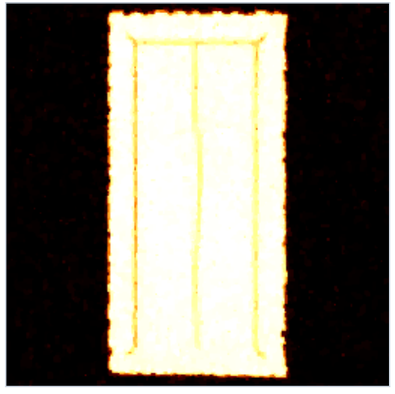
GS-PnP GS-DRUNet tuned



PSNR 33.96 | SSIM 0.892

HaarPSI 0.906 | MSE 4.02e-04

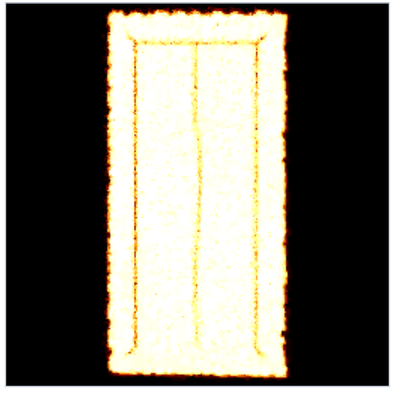
TV tuned (0.011)



PSNR 29.47 | SSIM 0.727

HaarPSI 0.767 | MSE 1.13e-03

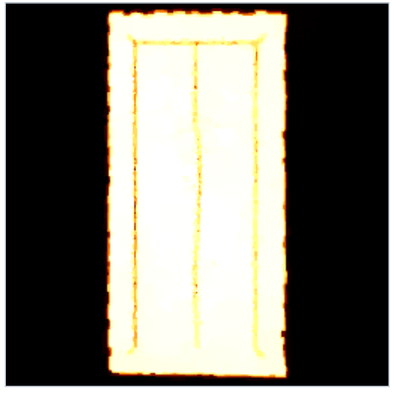
Old clean U-TV



PSNR 28.18 | SSIM 0.810

HaarPSI 0.764 | MSE 1.52e-03

New U-TV C60



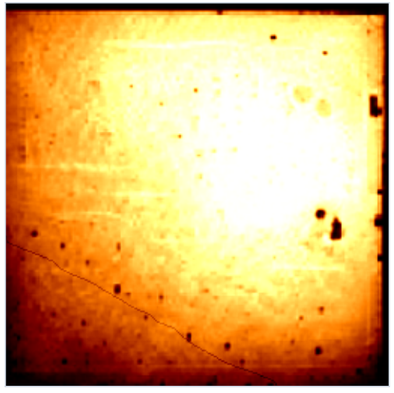
PSNR 31.32 | SSIM 0.948

HaarPSI 0.857 | MSE 7.37e-04

Perovskite | image 0031 | 20% PCM / NPL default

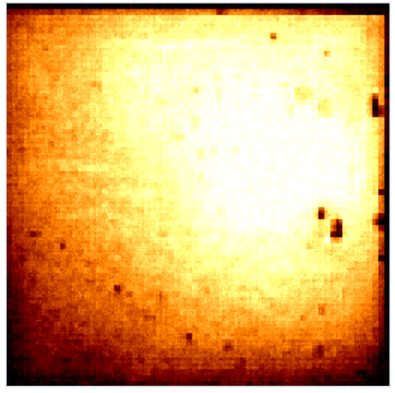
Review small dots, fine cracks, white gaps, and staircase artefacts.

Ground truth



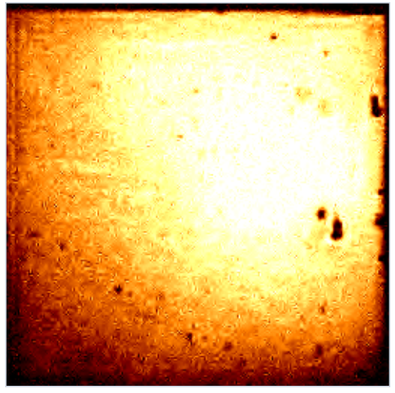
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



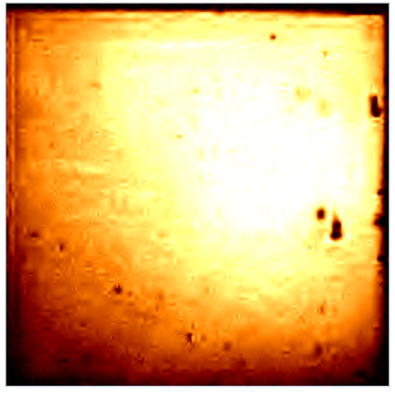
PSNR 27.19 | SSIM 0.606
HaarPSI 0.616 | MSE 1.91e-03

SPGL1-BP



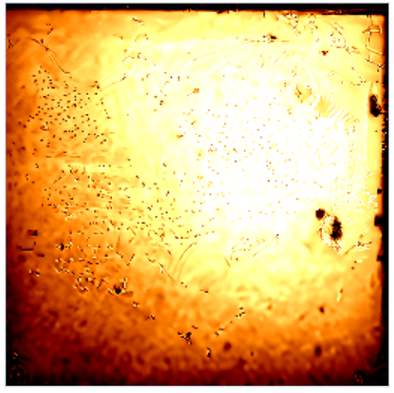
PSNR 25.94 | SSIM 0.505
HaarPSI 0.654 | MSE 2.55e-03

SPGL1-BPDN



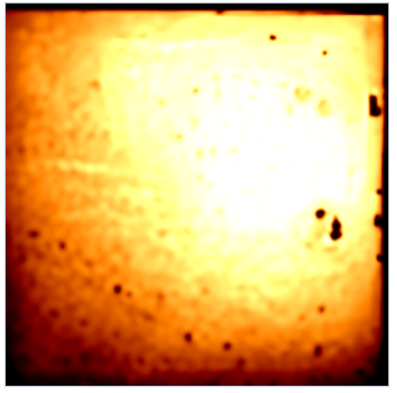
PSNR 29.82 | SSIM 0.751
HaarPSI 0.685 | MSE 1.04e-03

PnP-ADMM DRUNet



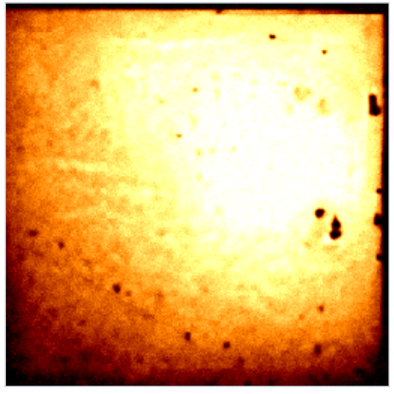
PSNR 23.87 | SSIM 0.558
HaarPSI 0.702 | MSE 4.10e-03

PnP-ADMM DRUNet tuned



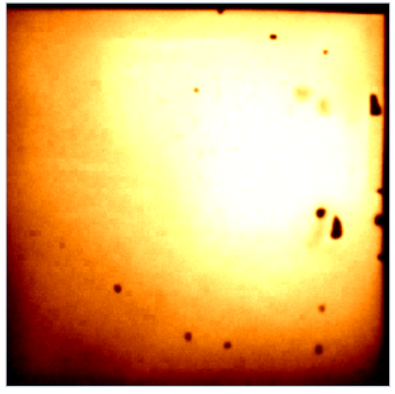
PSNR 33.00 | SSIM 0.849
HaarPSI 0.789 | MSE 5.01e-04

GS-PnP GS-DRUNet



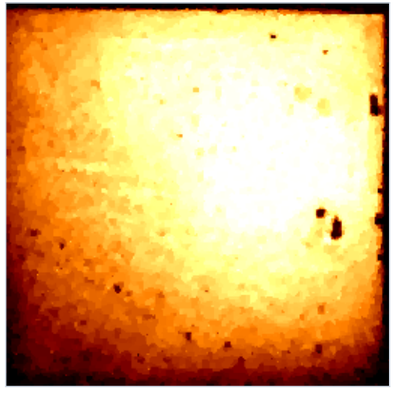
PSNR 30.10 | SSIM 0.699
HaarPSI 0.741 | MSE 9.77e-04

GS-PnP GS-DRUNet tuned



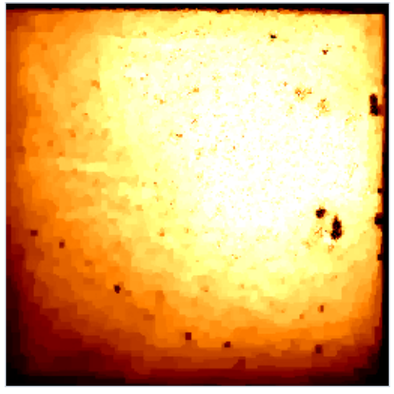
PSNR 31.14 | SSIM 0.785
HaarPSI 0.732 | MSE 7.69e-04

TV tuned (0.011)



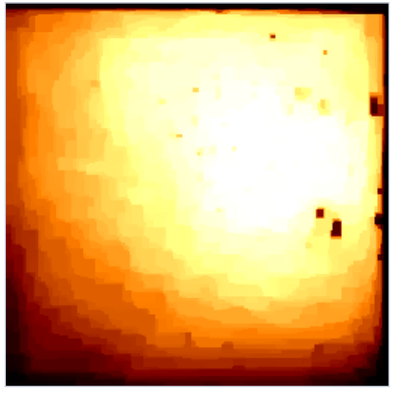
PSNR 30.68 | SSIM 0.773
HaarPSI 0.745 | MSE 8.56e-04

Old clean U-TV



PSNR 28.22 | SSIM 0.698
HaarPSI 0.730 | MSE 1.51e-03

New U-TV C60

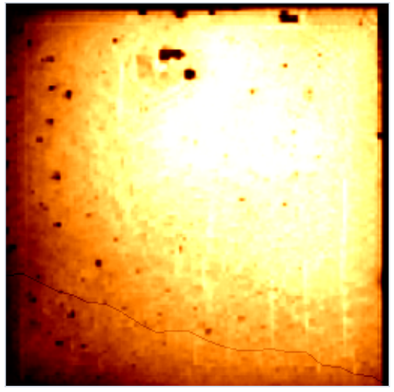


PSNR 30.46 | SSIM 0.780
HaarPSI 0.615 | MSE 9.00e-04

Perovskite | image 0079 | 20% PCM / NPL default

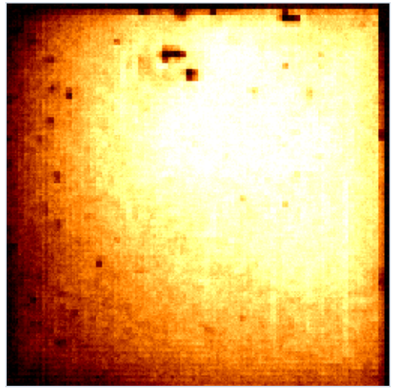
Review small dots, fine cracks, white gaps, and staircase artefacts.

Ground truth



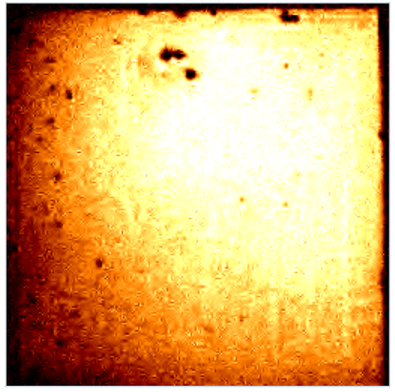
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



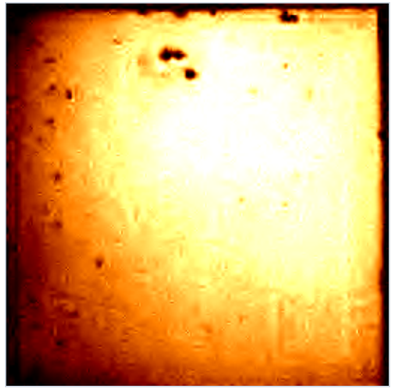
PSNR 27.03 | SSIM 0.608
HaarPSI 0.618 | MSE 1.98e-03

SPGL1-BP



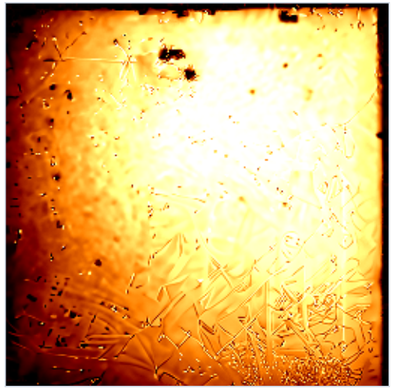
PSNR 25.72 | SSIM 0.508
HaarPSI 0.629 | MSE 2.68e-03

SPGL1-BPDN



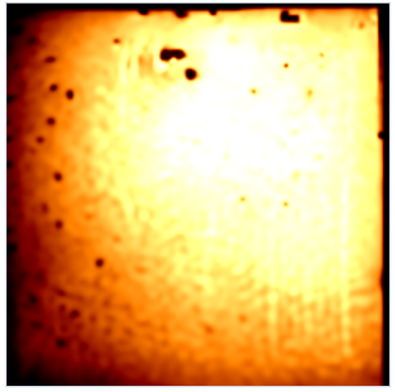
PSNR 29.42 | SSIM 0.746
HaarPSI 0.661 | MSE 1.14e-03

PnP-ADMM DRUNet



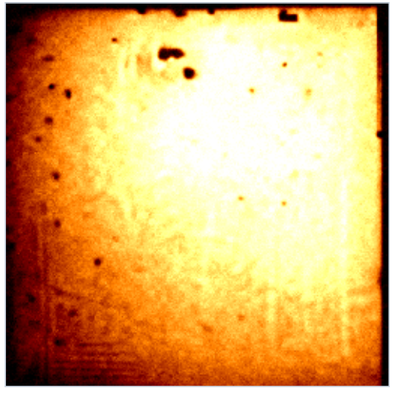
PSNR 23.80 | SSIM 0.560
HaarPSI 0.707 | MSE 4.17e-03

PnP-ADMM DRUNet tuned



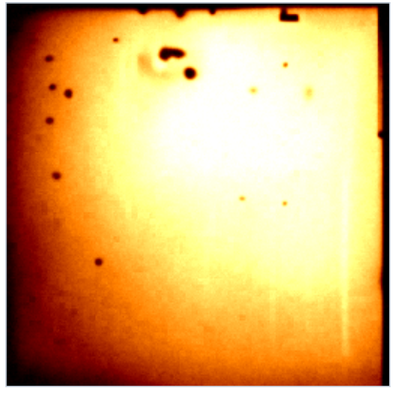
PSNR 32.73 | SSIM 0.846
HaarPSI 0.787 | MSE 5.33e-04

GS-PnP GS-DRUNet



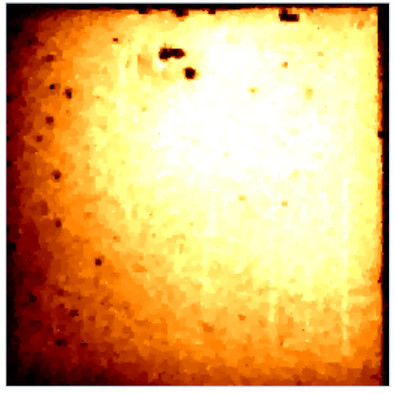
PSNR 29.85 | SSIM 0.697
HaarPSI 0.726 | MSE 1.04e-03

GS-PnP GS-DRUNet tuned



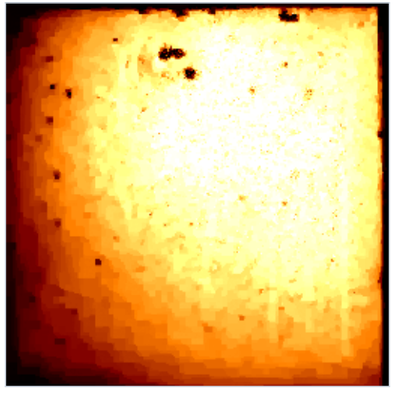
PSNR 31.14 | SSIM 0.782
HaarPSI 0.737 | MSE 7.69e-04

TV tuned (0.011)



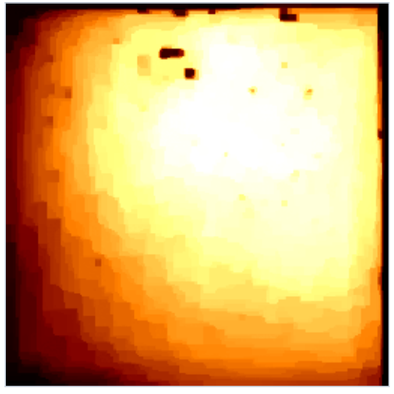
PSNR 30.44 | SSIM 0.771
HaarPSI 0.729 | MSE 9.04e-04

Old clean U-TV



PSNR 28.11 | SSIM 0.691
HaarPSI 0.727 | MSE 1.55e-03

New U-TV C60



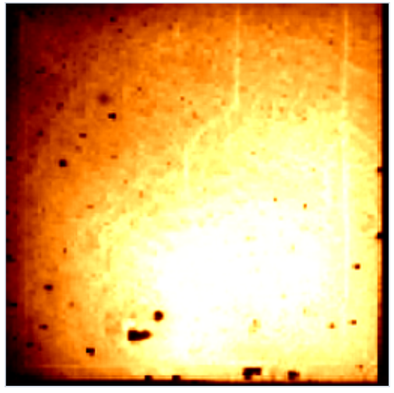
PSNR 29.98 | SSIM 0.762
HaarPSI 0.613 | MSE 1.00e-03

 | Every panel uses the same noisy acquisition. Shared AFM colormap, fixed [0,1] intensity scale. Metrics from paired 450-case test.

Perovskite | image 0098 | 20% PCM / NPL default

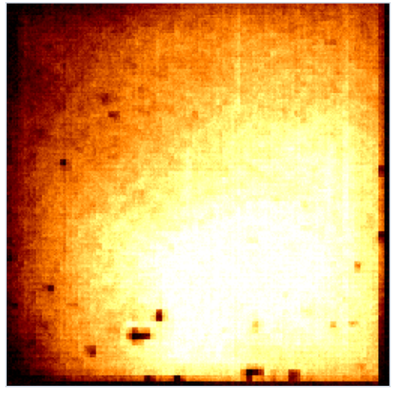
Review small dots, fine cracks, white gaps, and staircase artefacts.

Ground truth



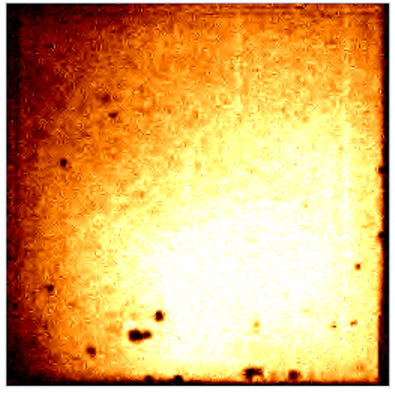
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



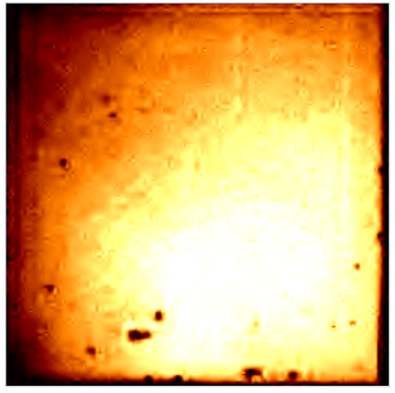
PSNR 27.22 | SSIM 0.608
HaarPSI 0.624 | MSE 1.90e-03

SPGL1-BP



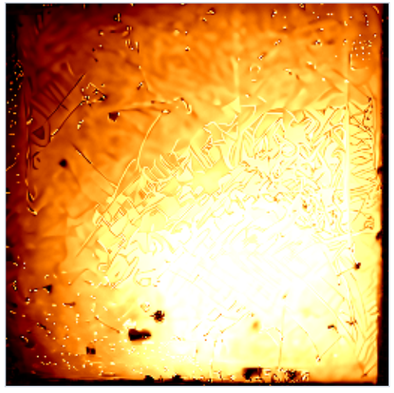
PSNR 25.89 | SSIM 0.506
HaarPSI 0.641 | MSE 2.58e-03

SPGL1-BPDN



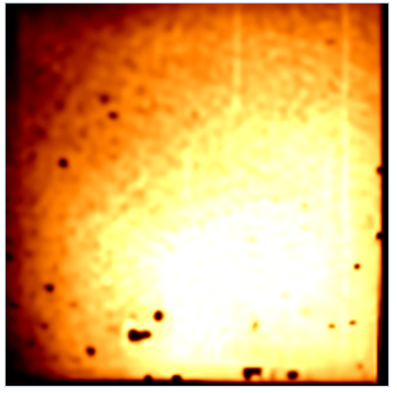
PSNR 29.65 | SSIM 0.755
HaarPSI 0.646 | MSE 1.08e-03

PnP-ADMM DRUNet



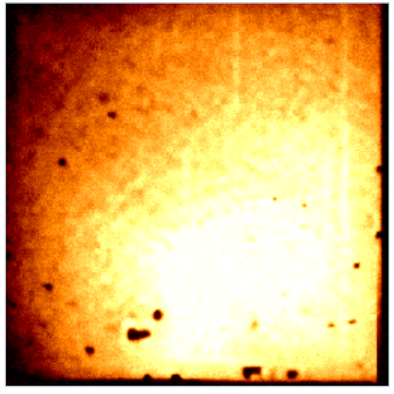
PSNR 23.85 | SSIM 0.529
HaarPSI 0.684 | MSE 4.12e-03

PnP-ADMM DRUNet tuned



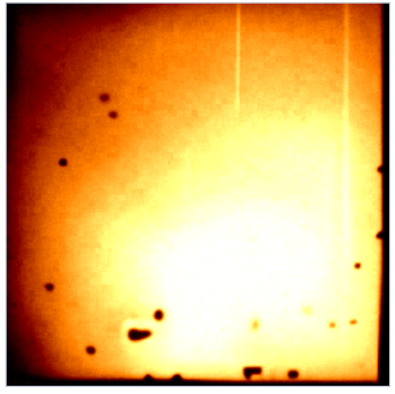
PSNR 32.91 | SSIM 0.854
HaarPSI 0.785 | MSE 5.11e-04

GS-PnP GS-DRUNet



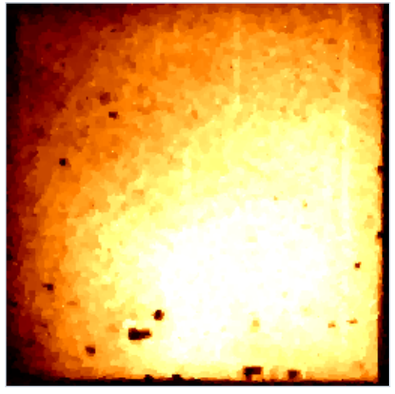
PSNR 30.17 | SSIM 0.704
HaarPSI 0.736 | MSE 9.61e-04

GS-PnP GS-DRUNet tuned



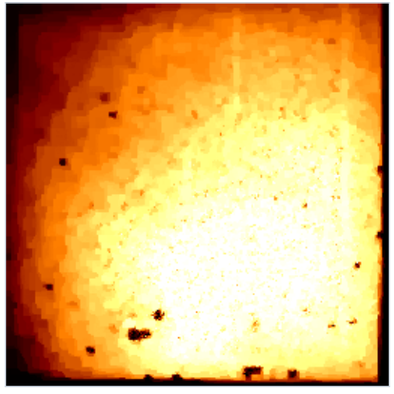
PSNR 31.28 | SSIM 0.792
HaarPSI 0.734 | MSE 7.45e-04

TV tuned (0.011)



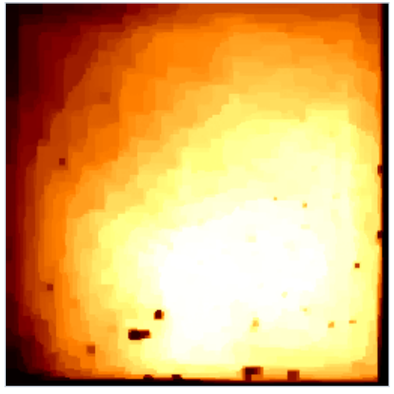
PSNR 30.72 | SSIM 0.781
HaarPSI 0.729 | MSE 8.48e-04

Old clean U-TV



PSNR 28.50 | SSIM 0.707
HaarPSI 0.730 | MSE 1.41e-03

New U-TV C60

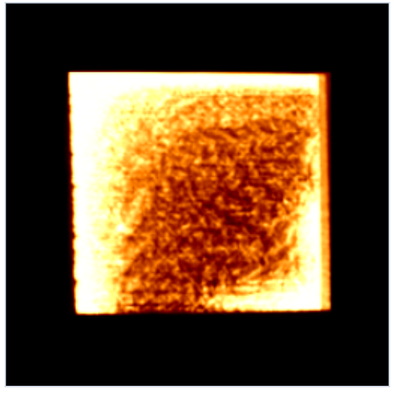


PSNR 30.46 | SSIM 0.780
HaarPSI 0.616 | MSE 9.00e-04

Perovskite | image 0260 | 20% PCM / NPL default

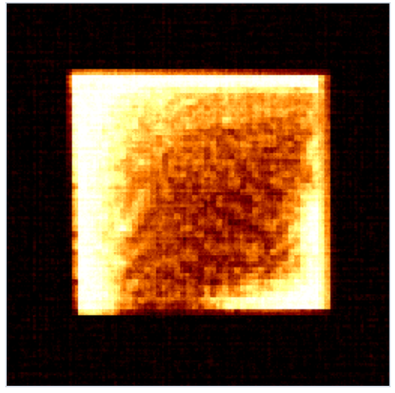
Review small dots, fine cracks, white gaps, and staircase artefacts.

Ground truth



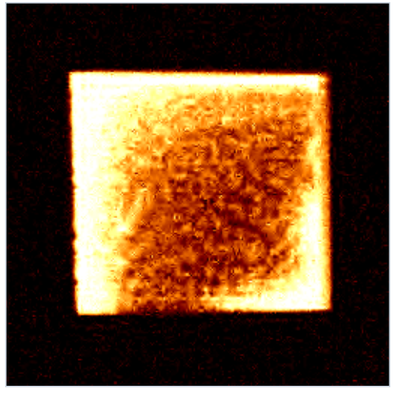
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



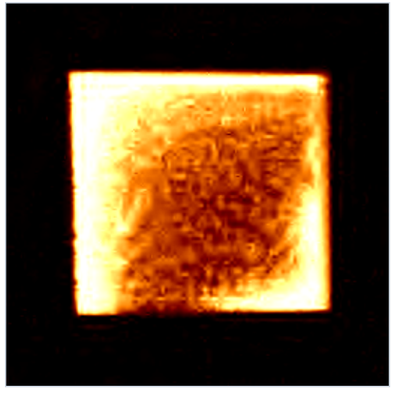
PSNR 24.51 | SSIM 0.473
HaarPSI 0.532 | MSE 3.54e-03

SPGL1-BP



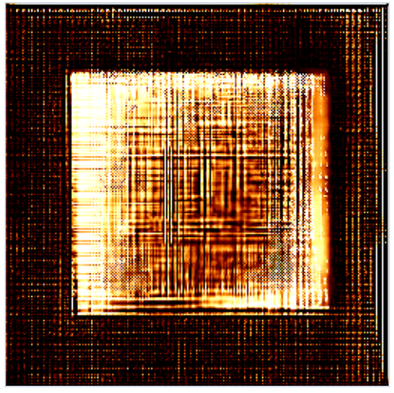
PSNR 25.31 | SSIM 0.467
HaarPSI 0.594 | MSE 2.95e-03

SPGL1-BPDN



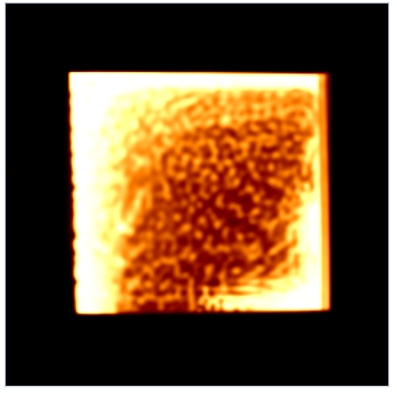
PSNR 27.66 | SSIM 0.709
HaarPSI 0.580 | MSE 1.71e-03

PnP-ADMM DRUNet



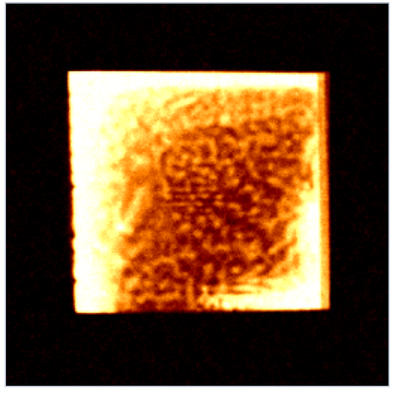
PSNR 7.90 | SSIM 0.091
HaarPSI 0.277 | MSE 1.62e-01

PnP-ADMM DRUNet tuned



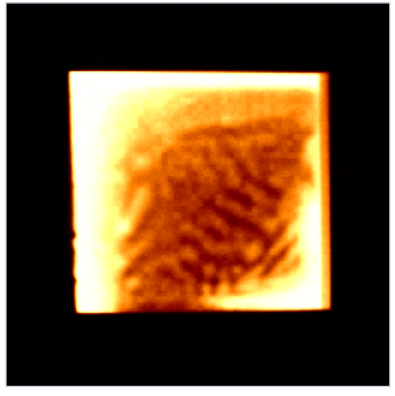
PSNR 32.27 | SSIM 0.928
HaarPSI 0.815 | MSE 5.92e-04

GS-PnP GS-DRUNet



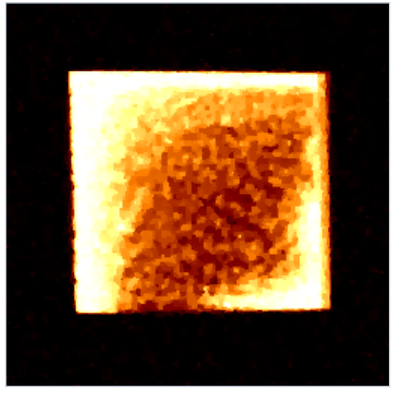
PSNR 29.89 | SSIM 0.645
HaarPSI 0.780 | MSE 1.03e-03

GS-PnP GS-DRUNet tuned



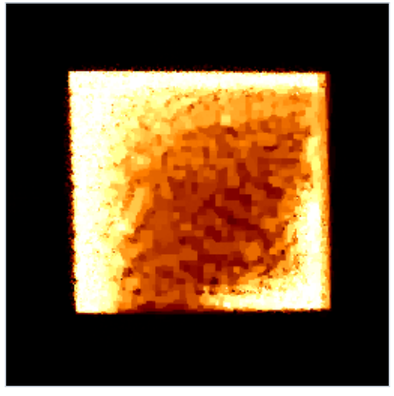
PSNR 30.04 | SSIM 0.800
HaarPSI 0.744 | MSE 9.92e-04

TV tuned (0.011)



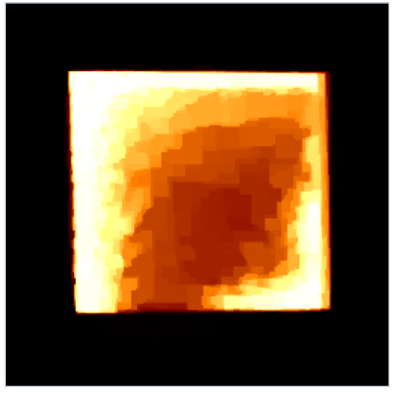
PSNR 29.79 | SSIM 0.731
HaarPSI 0.748 | MSE 1.05e-03

Old clean U-TV



PSNR 27.68 | SSIM 0.828
HaarPSI 0.679 | MSE 1.71e-03

New U-TV C60

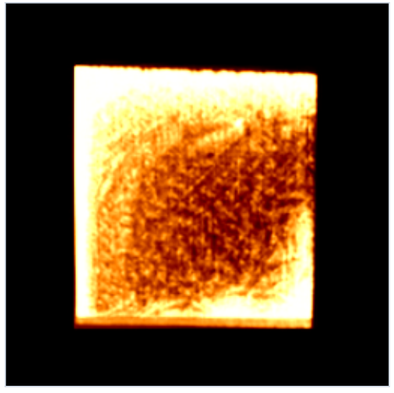


PSNR 28.53 | SSIM 0.806
HaarPSI 0.554 | MSE 1.40e-03

Perovskite | image 0288 | 20% PCM / NPL default

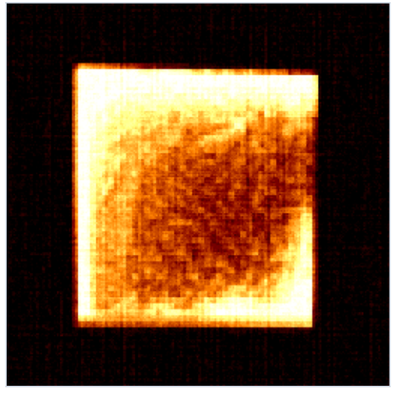
Review small dots, fine cracks, white gaps, and staircase artefacts.

Ground truth



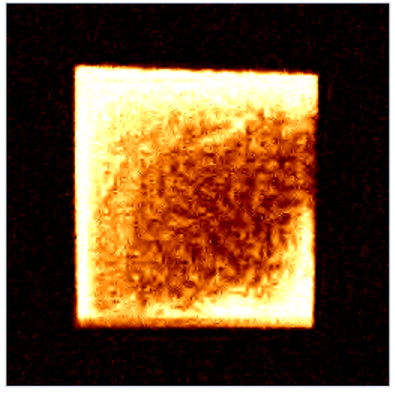
PSNR inf | SSIM 1.000
HaarPSI 1.000 | MSE 0

Zero-filled



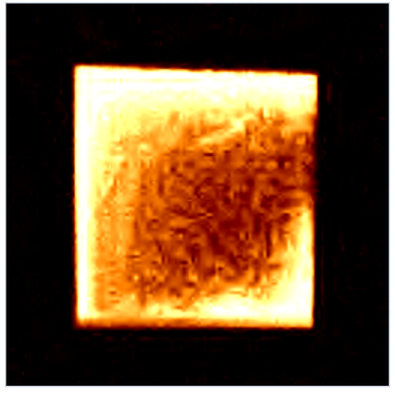
PSNR 24.69 | SSIM 0.464
HaarPSI 0.544 | MSE 3.40e-03

SPGL1-BP



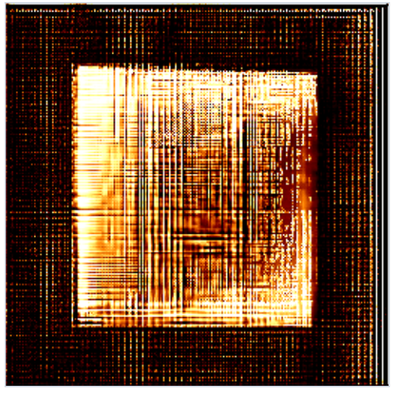
PSNR 25.64 | SSIM 0.460
HaarPSI 0.630 | MSE 2.73e-03

SPGL1-BPDN



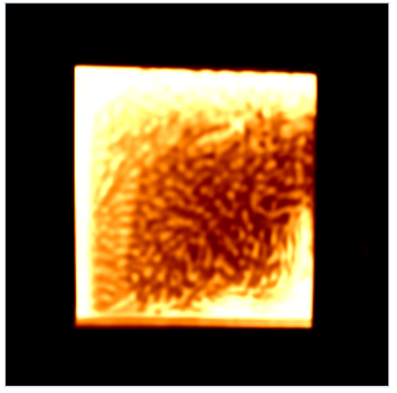
PSNR 28.27 | SSIM 0.706
HaarPSI 0.605 | MSE 1.49e-03

PnP-ADMM DRUNet



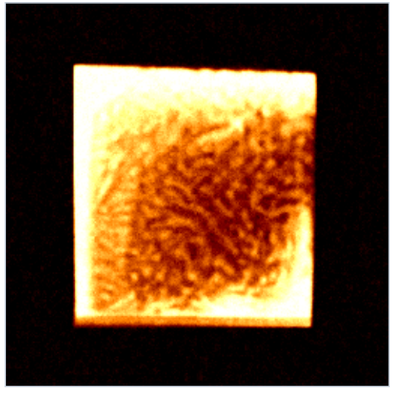
PSNR 6.78 | SSIM 0.091
HaarPSI 0.252 | MSE 2.10e-01

PnP-ADMM DRUNet tuned



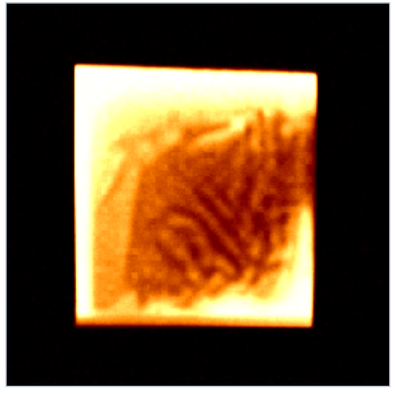
PSNR 32.44 | SSIM 0.927
HaarPSI 0.818 | MSE 5.70e-04

GS-PnP GS-DRUNet



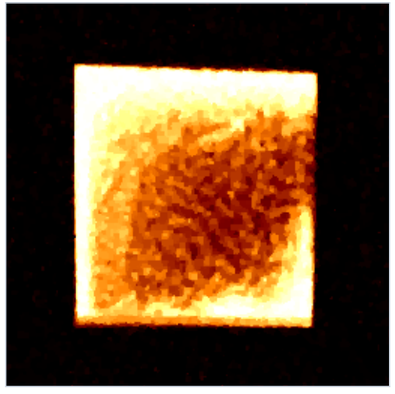
PSNR 29.93 | SSIM 0.641
HaarPSI 0.773 | MSE 1.02e-03

GS-PnP GS-DRUNet tuned



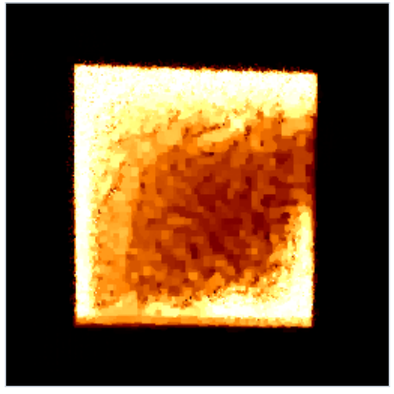
PSNR 30.03 | SSIM 0.798
HaarPSI 0.738 | MSE 9.92e-04

TV tuned (0.011)



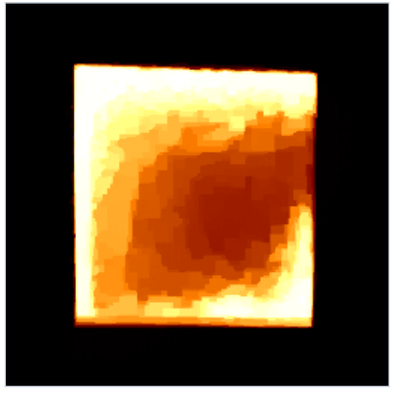
PSNR 29.88 | SSIM 0.726
HaarPSI 0.744 | MSE 1.03e-03

Old clean U-TV



PSNR 27.72 | SSIM 0.823
HaarPSI 0.682 | MSE 1.69e-03

New U-TV C60



PSNR 28.70 | SSIM 0.803
HaarPSI 0.581 | MSE 1.35e-03

 | Every panel uses the same noisy acquisition. Shared AFM colormap, fixed [0,1] intensity scale. Metrics from paired 450-case test.